COURSE PLAN

FIRST: BASIC INFORMATION

College	
College	: Prince Abdullah bin Ghazi Faculty of Information Technology
Department	: Computer Science
Course	
Course Title	: Theory of Computation and automata
Course Code	: 30801210
Credit Hours	: 3
Prerequisite	: -
Instructor	
Name	: Dr. Hasan Rashaideh
Office No.	: 2nd Floor (CS office - 502)
Tel (Ext)	: -
E-mail	: rashaideh@gmail.com
Office Hours	: Sun-Tue-Thu. (12:00 – 01:00pm)

Class Times

Building	Day	Start Time	End Time	Room No.
IT	Sun-Tue-Thu.	8:00	9:00	

Text Book	
Title	: An Introduction to Formal Languages and Automata, 3rd Edition, By Peter Linz, Jones and Publishers, 2001.

References

1- Introduction to computer theory, 2nd edition, by Daniel I. A. Cohen, Wiley, 1997

SECOND: PROFESSIONAL INFORMATION

COURSE DESCRIPTION

This course introduces the theory of computation through a set of abstract machines that serve as models for computation - finite automata, pushdown automata, and Turing machines – and examines the relationship between these automata and formal languages. Additional topics beyond the automata classes themselves include deterministic and nondeterministic machines, regular expressions, context free grammars, undesirability, and the P = NP question.

COURSE OBJECTIVES

1. Introduce concepts in automata theory and theory of computation
2. Identify different formal language classes and their relationships
3. Design grammars and recognizers for different formal languages
4. Prove or disprove theorems in automata theory using its properties.

COURSE LEARNING OUTCOMES

1) **Knowledge and Understanding**
   - Acquire a full understanding and mentality of Automata Theory as the basis of all computer science languages design
   - Have a clear understanding of the Automata theory concepts such as RE's, DFA's, NFA's, Stack's, Turing machines, and Grammars.
   - Design FAs, NFAs, Grammars, languages modelling, small compilers basic
   - Minimize FA's and Grammars of Context Free Languages
2) **Professional Skills**
   - to recognize the significance of complexity classes and analysis and to deduce the complexity of certain types of algorithm
   - to demonstrate a deeper and broader understanding of classes of complexity
   - to prove the basic results of the Theory of Computation
3) **Competences (Transferable skill and attributes)**
   - Search for, analysis and synthesis of data and information, with the use of the necessary technology.

COURSE SYLLABUS

Week	Course Topic	Notes
Week 1	**Introduction To The Theory Of Computation.**	
Week 2	**Deterministic Finite Automata.**	
Week 3	**Nondeterministic Finite Automata**	
Week 4	**Regular Expressions** **Connection Between Regular Expression and Regular Languages.**	
Week 5	**Regular Grammars**	
Week 6	**Closure Properties of Regular Languages.**	
Week 7	**Elementary Questions about Regular Languages**	
Week 8		**Midterm Exam**
Week 9	**Identifying Non regular Languages.**	
Week 10	**Context Free Grammars (CFGs).** **Parsing and Ambiguity.**	
Week 11	**Context Free Grammars and Programming Languages.**	
Week 12	**Methods for Transforming Grammars.**	
Week 13	**Two Important Normal Forms**	
Week 14	**Nondeterministic Pushdown Automata** **Pushdown Automata And CFGs**	
Week 15	**Deterministic Pushdown Automata And Deterministic CFLs.**	
Week 16	**Final Exam**	**Final Exam**

COURSE LEARNING RESOURCES

Teaching materials for students (handouts, exercise collections, examples of solved exercises), detailed course syllabus, and all other data are available by e-learning MOODLE system to the registered students via elearning.bau.edu.jo.

ONLINE RESOURCES

- NA

ASSESSMANT TOOLS

(Write assessment tools that will be used to test students ability to understand the course material and gain the skills and competencies stated in learning outcomes

ASSESSMENT TOOLS	%
Quizzes	5
Assignments	5
Mid Exam	40
Final Exam	50
TOTAL MARKS	100

THIRD: COURSE RULES

ATTENDANCE RULES

Attendance and participation are extremely important, and the usual University rules will apply. Attendance will be recorded for each class. Absence of 10% will result in a first written warning. Absence of 15% of the course will result in a second warning. Absence of 20% or more will result in forfeiting the course and the student will not be permitted to attend the final examination. Should a student encounter any special circumstances (i.e. medical or personal), he/she is encouraged to discuss this with the instructor and written proof will be required to delete any absences from his/her attendance records.

GRADING SYSTEM
Example:

90 – 100	**A**
85 – 89	**B+**
75 – 84	**B**
65 – 74	**C+**
60 – 64	**C**
55 – 59	**D+**
50 – 54	**D**
45-49	**D-**
Less than 45	**F**

REMARKS

Attendance is required for this class. Participation of in-class discussions and activities is also required. All submitted assignments and projects must be done by the student(s). It is a violation of the university regulations to submit other's work and the instructor of this course takes the violations very seriously.

COURSE COORDINATOR

Course Coordinator: Dr. Hasan Rashaideh	**Department Head: Dr. Omar Alzubi**
Signature:	**Signature:**
Date:	**Date:**